DECOMPOSER BIOLOGY · CROP RESIDUE AFTER HARVEST

MYCRO-TEC®

DIGESTER

Decomposer biology for post-harvest residue.



Four selected organisms applied once onto stubble, prunings or crop trash after harvest, with an ammonium sulphate starter, timed to moisture. They begin breaking down the structural fibre so the carbon and nutrients in the residue return to the soil ahead of the next crop.



VEGETABLES



ORCHARDS



BROADACRE

Application

CROP GROUP	DIGESTER	STARTER	WATER
Vegetables Crop residue after harvest	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
Orchards Prunings and leaf litter on the orchard floor. Not for nut trees.	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
Broadacre Cereal, canola and legume stubble	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. Applied once, after harvest, onto the residue. Not for in-crop use.

What is in it

Trichoderma harzianum FUNGUS	$>2\times10^{8}$ CFU/ML	Fast fungal coloniser of cellulose-rich residue; its enzymes begin opening the fibre.
Aspergillus niger FUNGUS	$>2\times10^{7}$ CFU/ML	Produces organic acids and enzymes that work on the tougher, more lignified material.
Streptomyces lydicus ACTINOMYCETE	$>2\times10^{7}$ CFU/ML	Adds bacterial decomposition alongside the fungi, so breakdown does not rely on one pathway.
Penicillium radicum FUNGUS	$>1\times10^{7}$ CFU/ML	Broadens the decomposer team across the residue surface and the damp base.

Colony-forming units per millilitre at production, as declared on the label. Three fungi and one actinomycete. They act on cellulose, hemicellulose, lignin and pectin.

MIXING ORDER

01
Shake the drum. Add Digester to cool, clean, non-chlorinated water first.

02
Add 8 kg/ha ammonium sulphate and stir thoroughly. Strain as you pour; filters 150 µm or larger.

03
Apply the same day onto the residue. Do not leave it in the tank overnight.

When Digester is relevant

SITUATION	WHY IT MATTERS
Residue that will still be there at sowing	Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.
Heavy or lignified loads	Big stubble loads, or material that has sat through a dry summer and stalled.
A moisture window you can hit	Applied soon after harvest and close to forecast rain. The breakdown runs on residue moisture.
Where the basics are already in hand	Residue breaks down fastest in a block where water, soil balance and nutrition are being managed. Digester is a step in that program, not a substitute for it.

What slows it down. Hot, dry spells. Activity falls away through prolonged heat and dryness. Line the application up with a viable moisture window. **Time in the tank.** Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles. **Nitrogen on its own.** Nitrogen feeds whatever biology is already there. It does not add the decomposers. The ammonium sulphate is a starter, not the treatment.

How you check it. Digestion starts at the stubble base, where moisture sits. Treated residue pulls away from the soil more easily and the stalks soften and weaken as it runs. That is the one field check a grower can make without a lab.

What to expect

SAY	NOT A CLAIM WE MAKE
Supports biological breakdown of crop residue.	Removes all residue, or guarantees it will disappear.
Acts on the structural parts of residue: cellulose, hemicellulose, lignin and pectin.	Controls stubble disease. Competition for residue is not a disease position.
One application after harvest, timed to moisture.	Works in-crop, on nut trees, or without the starter.

Storage

Store at 5–40 °C, out of direct sunlight, drums upright. The lids are vented. Do not freeze.

Twelve months from the date of manufacture. Use opened drums promptly rather than keeping them.

Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

Safety

Keep out of reach of children. May irritate skin or eyes; wash after handling.

Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.

Not certified organic. Allowable-input status is assessed case by case with your certifier.

If you run a nutrition program with us, Digester is the post-harvest step and the prescription blend at establishment follows it. If you don't, it stands on its own after harvest.

Talk to us about Digester

Tell us the crop, the residue load and when you expect rain. Your Hybrid-Ag agronomist can advise on timing and rate.

(03) 5722 7555 **hybridag.com.au**

Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.

